WORKED SOLUTION

STEP II 1995, Question 11

Mechanics • Linear Air Resistance

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Read the question before you go on. It is in the STEP Support Programme's question database:

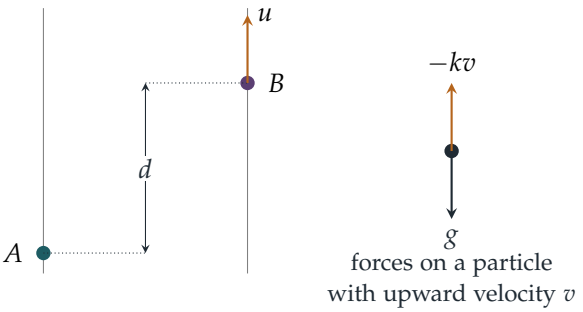
step.maths.org/questions/95-s2-q11

The key idea

Because the resistance is *linear* in velocity, the equation of motion is linear, and the same equation holds whether a particle is rising or falling. So subtract the two equations of motion: gravity cancels, and the *relative* velocity simply decays exponentially. Integrating that once gives the separation directly, without ever needing to know where either particle is.

Solution

Model each body as a particle of unit mass moving on a vertical line, acted on by its weight g and a resistance of magnitude $k|v|$ directed against its motion. Take the upward direction as positive, put the origin at A, and let particle 1 (from A) and particle 2 (from B) have heights x_1, x_2 and velocities v_1, v_2 at time t.



Equations of motion

For a particle with upward velocity v, the resistance is $-kv$: if $v > 0$ (rising) it points down, and if $v < 0$ (falling) it points up, which is exactly "against the motion" in both

cases. So Newton's second law for unit mass gives, for each particle,

$$\frac{\mathrm{d}v_1}{\mathrm{d}t} = -g - kv_1, \qquad \frac{\mathrm{d}v_2}{\mathrm{d}t} = -g - kv_2,$$

valid for all $t \geq 0$.

Ask yourself Particle 2 goes up, stops and comes down. Don't we need a separate equation for each stage?

With a resistance proportional to *speed squared* we would: the force would be $-kv|v|$, which changes form when v changes sign. Here the force is $-kv$, a single expression that already reverses direction when v does, so one equation covers the whole motion.

The relative motion

Let $w = v_2 - v_1$ be the velocity of particle 2 relative to particle 1. Subtracting the two equations, the weight terms cancel:

$$\frac{\mathrm{d}w}{\mathrm{d}t} = -kw, \qquad w(0) = u - 0 = u,$$

so $w = u\,\mathrm{e}^{-kt}$. Now let $s = x_2 - x_1$, with $s(0) = d$ since B is a height d above A. Then $\mathrm{d}s/\mathrm{d}t = w$, and

$$s = d + \int_0^t u\,\mathrm{e}^{-k\tau}\,\mathrm{d}\tau = d + \frac{u}{k}\left(1 - \mathrm{e}^{-kt}\right).$$

Since $u \geq 0$ and $1 - \mathrm{e}^{-kt} \geq 0$, we have $s \geq d > 0$: particle 2 is always above particle 1, and the particles never meet. So the distance between them at time t is

$$\boxed{d + \frac{u}{k}\left(1 - \mathrm{e}^{-kt}\right)}$$

As $t \to \infty$, $\mathrm{e}^{-kt} \to 0$ because $k > 0$, so the distance tends to $d + \dfrac{u}{k}$, as required.

The individual motions (if you prefer to go the long way)

Solving each equation separately (it is first-order linear, or separable) gives $v = -\frac{g}{k} + \left(v(0) + \frac{g}{k}\right)\mathrm{e}^{-kt}$, and integrating again,

$$x_1 = -\frac{gt}{k} + \frac{g}{k^2}\left(1 - \mathrm{e}^{-kt}\right),$$
$$x_2 = d - \frac{gt}{k} + \frac{1}{k}\left(u + \frac{g}{k}\right)\left(1 - \mathrm{e}^{-kt}\right).$$

Subtracting, the gt/k and g/k^2 terms cancel and we recover the same s. Both particles approach the terminal velocity $-g/k$, which is why their separation settles down.

Checks

- At $t = 0$ the distance is d; and for small t it is about $d + ut$, as it would be without resistance (both particles then fall equally under gravity).
- If $u = 0$ the particles are released together from rest, follow identical motions, and stay d apart, as the formula says.
- Units: k has units s^{-1} (force per unit mass per unit velocity), so u/k is a length.
- Numerically, stepping both equations of motion with a Runge–Kutta method for random k, u, d, t matched the formulas to about 10^{-12}. With $k = 0.5$, $u = 10$, $d = 3$, the distance is 10.87 at $t = 1$ and 22.999 at $t = 20$, approaching $d + u/k = 23$.

Common slips

- Splitting particle 2's motion into "up" and "down" phases with different signs for the resistance: unnecessary here, and a frequent source of sign errors.
- Writing the resistance as $+kv$ for the falling particle when v is already the (negative) upward velocity, which double-counts the sign.

Beyond STEP

The trick of subtracting the equations works because the equation of motion is linear: the difference of two solutions solves the homogeneous equation $\dot{w} = -kw$. This is the superposition principle that organises the whole theory of linear differential equations, where every solution is "particular solution plus complementary function". With quadratic drag the principle fails and the separation behaves quite differently. Boyce and DiPrima, *Elementary Differential Equations*, develops the linear theory; Kleppner and Kolenkow, *An Introduction to Mechanics*, treats motion with both kinds of resistance.

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